Rotational Dynamics – Formula Sheet

Physics, Class 12, Chapter 1 · Maharashtra Board (HSC)

Circular Motion

$$a_c = \frac{v^2}{r} = r\omega^2 = v\omega = \frac{4\pi^2 r}{T^2} \qquad F_c = \frac{mv^2}{r} = mr\omega^2 \qquad \omega = \frac{2\pi N}{60} = \frac{2\pi}{T} = 2\pi n$$

Vertical Circular Motion

$$T_{\text{top}} = \frac{mv_t^2}{r} - mg \qquad T_{\text{bot}} = \frac{mv_b^2}{r} + mg \qquad T_{\text{bot}} - T_{\text{top}} = 6mg$$

$$v_{\text{top (min)}} = \sqrt{gr} \qquad v_{\text{bot (min)}} = \sqrt{5gr} \qquad v_b^2 = v_t^2 + 4gr$$

Vehicles on Circular Roads

Flat road: $v_{\max} = \sqrt{\mu_s r g}$ Banked: $\tan\theta = \dfrac{v^2}{rg}$ Banked + friction: $v_{\max} = \sqrt{\dfrac{rg(\mu_s + \tan\theta)}{1 - \mu_s \tan\theta}}$

Rotational Kinematics

$$v = r\omega \qquad a_t = r\alpha \qquad a_c = r\omega^2$$
$$\omega = \omega_0 + \alpha t \qquad \theta = \omega_0 t + \tfrac{1}{2}\alpha t^2 \qquad \omega^2 = \omega_0^2 + 2\alpha\theta$$

Torque, Work, Power

$$\vec{\tau} = \vec{r} \times \vec{F}, \;\; \tau = rF\sin\theta \qquad \tau_{\text{couple}} = F \times \text{arm} \qquad \tau = I\alpha$$
$$W = \tau\theta \qquad P = \tau\omega \qquad \tau = \frac{dL}{dt}$$

Moment of Inertia

$I = \sum m_i r_i^2 = \int r^2 dm = MK^2$ Parallel: $I = I_c + Mh^2$ Perpendicular (lamina): $I_z = I_x + I_y$

Body	I	K^2/R^2
Ring / hollow cylinder (central axis)	MR^2	1
Disc / solid cylinder	$\frac{1}{2}MR^2$	1/2
Hollow sphere	$\frac{2}{3}MR^2$	2/3
Solid sphere	$\frac{2}{5}MR^2$	2/5
Rod about centre	$\frac{ML^2}{12}$	—
Rod about end	$\frac{ML^2}{3}$	—

Angular Momentum

$L = I\omega$ (rigid body) $L = mvr$ (particle) Conservation: $I_1\omega_1 = I_2\omega_2$ ($\tau_{\text{ext}} = 0$)

Rolling Motion

Pure rolling: $v = r\omega$ $KE_{\text{total}} = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2\left(1 + \dfrac{K^2}{R^2}\right)$

On incline: $v = \sqrt{\dfrac{2gh}{1 + K^2/R^2}}$ $a = \dfrac{g\sin\theta}{1 + K^2/R^2}$

Race order: solid sphere > disc > hollow sphere > ring (frictionless sliding: $a = g\sin\theta$, beats all rollers)

Linear ↔ Rotational Analogy

s	θ	$p = mv$	$L = I\omega$
v, a	ω, α	$F = ma$	$\tau = I\alpha$
m	I	$W = Fs$	$W = \tau\theta$
F	τ	$KE = \frac{1}{2}mv^2$	$KE = \frac{1}{2}I\omega^2$

Quick Constants & Facts

- 1 revolution $= 2\pi$ rad $= 360°$; $\quad 1$ rad $\approx 57.3°$
- Centripetal force is a requirement, not a new force (tension / friction / gravity supply it)
- Banking angle is mass-independent; designed speed same for all vehicles
- I minimum about centre-of-mass axis; only static friction enables rolling but does no work
- Board weightage: 5 marks (7 with option)